

EE 5239 Homework 5 Solution

3.3.1 (4.3.1)

Bertsekas 3.3.1: The objective function is $(x - a)^2 + (y - b)^2 + xy$. The Hessian matrix corresponding to this objective function is $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ which is convex. Consequently, the global minimum occurs in the interior of the set $\{0 \leq x \leq 1, 0 \leq y \leq 1\}$ or on the boundary and this depends on the values of (a, b) . Taking derivatives w.r.t. x and y and setting them to zero, we get

$$\begin{aligned} 2(x - a) + y &= 0, \\ 2(y - b) + x &= 0. \end{aligned}$$

The solution is

$$x = \frac{4a - 2b}{3}, y = \frac{4b - 2a}{3}.$$

Since $\{0 \leq x \leq 1, 0 \leq y \leq 1\}$, $x = 0$ for $a \leq \frac{b}{2}$, $y = 0$ for $b \leq \frac{a}{2}$, $x = 1$ for $a \geq \frac{2b+3}{4}$ and $y = 1$ for $b \geq \frac{2a+3}{4}$.

3.3.3 (4.3.3)

Consider $N = 2$. Then the problem is

$$\max (1 + wu_1)(1 + wu_0) + (1 - u_1)(1 + wu_0) + (1 - u_0) \quad \text{s.t. } 0 \leq u_0, u_1 \leq 1.$$

The KKT conditions yield that

$$\begin{aligned} (1 - w) + (w - w^2)u_0 - \lambda_1^- + \lambda_1^+ &= 0, \\ (1 - 2w) + (w - w^2)u_1 - \lambda_0^- + \lambda_0^+ &= 0, \\ \lambda_0^- u_0 &= 0, \quad \lambda_0^+ (u_0 - 1) = 0, \quad \lambda_1^- u_1 = 0, \quad \lambda_1^+ (u_1 - 1) = 0, \end{aligned}$$

where $\lambda_0^\pm, \lambda_1^\pm \geq 0$.

- If $w > 1$, then $(1 - w) + (w - w^2)u_0 < 0$ and $(1 - 2w) + (w - w^2)u_1 < 0$, so $\lambda_0^+, \lambda_1^+ > 0$. This implies that $u_0 = u_1 = 1$.
- If $w < 1/2$, then $(1 - w) + (w - w^2)u_0 > 0$ and $(1 - 2w) + (w - w^2)u_1 > 0$, so $\lambda_0^-, \lambda_1^- > 0$. This implies that $u_0 = u_1 = 0$.
- If $1/2 < w < 1$, then $(1 - w) + (w - w^2)u_0 > 0$, so $\lambda_1^- > 0$. Then $(1 - 2w) + (w - w^2)u_1 < 0$, so $\lambda_0^+ > 0$. Hence, $u_0 = 1$ and $u_1 = 0$.
- If $w = 1/2$, then $\lambda_1^- > 0$. It follows that $\lambda_0^+ = \lambda_0^- = 0$. Plugging in $w = 1/2$ and $u_1 = 0$, the objective is reduced to a constant. Hence, $0 \leq u_0 \leq 1$ and $u_1 = 0$.
- If $w = 1$, then $\lambda_0^+ > 0$. It follows that $\lambda_1^+ = \lambda_1^- = 0$. Plugging in $w = 1$ and $u_0 = 1$, the objective is reduced to a constant. Hence, $u_0 = 1$ and $0 \leq u_1 \leq 1$.

3.4.1 (4.4.1)

The primal problem is

$$\begin{aligned} \max \quad & \sum_{i=1}^n p_i s_i \\ \text{s.t.} \quad & \sum_{i=1}^n s_i \leq Q, \\ & 0 \leq s_i \leq d_i, \quad i = 1, \dots, n. \end{aligned}$$

The dual problem is

$$\begin{aligned} \min \quad & \sum_{i=1}^n d_i \lambda_i + Qy \\ \text{s.t.} \quad & \lambda_i + y \geq p_i, \quad i = 1, \dots, n, \\ & \lambda_i \geq 0, \quad i = 1, \dots, n, \\ & y \geq 0. \end{aligned}$$

The slackness conditions are

$$\begin{aligned} \lambda_i(d_i - s_i) &= 0, \quad i = 1, \dots, n, \\ s_i(p_i - \lambda_i - y) &= 0, \quad i = 1, \dots, n. \end{aligned}$$

We know that the optimal primal and dual solutions should satisfy constraints for each problem and slackness conditions.

If $p_i > y$ then $\lambda_i > 0$ because of the constraint in the dual problem, $\lambda_i \geq p_i - y$. By the first slackness condition, we have $s_i = d_i$.

If $p_i < y$ then $s_i = 0$ because the second slackness condition, $p_i - \lambda_i - y$ cannot be 0.

If $p_i = y$ then the slackness conditions for i becomes $\lambda_i(d_i - s_i) = 0$ and $\lambda_i s_i = 0$. It means that s_i can be any value as long as it satisfies constraints in the primal problem.

Therefore, we can conclude that there exists a cutoff price level y . Using the result, we can get s_i^* by maximally assigning the amount of the item from the one with the highest price to the one with the lowest price until the total assignment reaches Q .

1. Sort items so that $p_1 \geq p_2 \geq \dots \geq p_n$.
2. Initialize $s_j \leftarrow 0$ for $j = 1, \dots, n$, $S \leftarrow Q$ and $i \leftarrow 1$.
3. Until $S = 0$, $s_i \leftarrow \min\{d_i, S\}$, $S \leftarrow S - s_i$ and $i \leftarrow i + 1$.

The procedure must terminate because $\sum_{i=1}^n d_i \geq Q$.

3.4.3 (4.4.3)

Let's first consider

$$(P) \quad \min_{A'x \geq b} c'x \iff \max_{A\mu = c, \mu \geq 0} b'\mu. \quad (D)$$

The dual problem to (P) is

$$\max_{\mu \geq 0} q(\mu) = \max_{\mu \geq 0} \inf_{x \in \mathbb{R}^n} \left\{ \sum_{j=1}^n \left(c_j - \sum_{i=1}^m \mu_i a_{ij} \right) x_j + \sum_{i=1}^m \mu_i b_i \right\}.$$

If $c_j - \sum_{i=1}^m \mu_i a_{ij} \neq 0$, then $q(\mu) = -\infty$. Thus the dual problem is

$$\begin{aligned} \max \sum_{i=1}^m \mu_i b_i \\ \sum_{i=1}^m \mu_i a_{ij} = c_j, \quad j = 1, \dots, n \\ \mu \geq 0. \end{aligned}$$

To find the dual of (D) , note that (D) is equivalent to

$$\min_{A\mu=c, \mu \geq 0} -b'\mu,$$

and so the dual problem is

$$\max_{x \in \mathbb{R}^n} p(x) = \max_x \inf_{\mu \geq 0} \{(Ax - b)'\mu - c'x\}.$$

If $a'_i x - b_i < 0$ for any i , then $p(x) = -\infty$. Thus the dual of (D) is

$$\begin{aligned} \max -c'x \quad \text{or} \quad \min c'x \\ \text{subject to } A'x \geq b. \end{aligned}$$

The Lagrangian optimality condition for (P) is

$$x^* = \arg \min_x \left\{ \left(c - \sum_{i=1}^m \mu_i^* a_i \right)' x + \sum_{i=1}^m \mu_i^* b_i \right\},$$

from which we determine the complementary slackness conditions for (P) :

$$A\mu = c.$$

The Lagrangian optimality condition for (D) is

$$\mu^* = \arg \min_{\mu \geq 0} \{(Ax^* - b)'\mu - c'x^*\},$$

from which we determine the complementary slackness conditions for (D) :

$$Ax^* - b \geq 0,$$

$$(Ax^* - b)_i \mu_i^* = 0, \quad \forall i.$$

Next, consider

$$(P) \quad \min_{A'x \geq b, x \geq 0} c'x \iff \max_{A\mu \leq c, \mu \geq 0} b'\mu. \quad (D)$$

The dual problem to (P) is

$$\max_{\mu \geq 0} q(\mu) = \max_{\mu \geq 0} \inf_{x \geq 0} \left\{ \sum_{j=1}^n \left(c_j - \sum_{i=1}^m \mu_i a_{ij} \right) x_j + \sum_{i=1}^m \mu_i b_i \right\}.$$

If $c_j - \sum_{i=1}^m \mu_i a_{ij} < 0$, then $q(\mu) = -\infty$. Thus the dual problem is

$$\begin{aligned} & \max \sum_{i=1}^m \mu_i b_i \\ & \sum_{i=1}^m \mu_i a_{ij} \leq c_j, \quad j = 1, \dots, n \\ & \mu \geq 0. \end{aligned}$$

To find the dual of (D), note that (D) is equivalent to

$$\min_{A\mu \leq c, \mu \geq 0} -b'\mu,$$

and so the dual problem is

$$\max_{x \geq 0} p(x) = \max_{x \geq 0} \inf_{\mu \geq 0} \{(Ax - b)'\mu - c'x\}.$$

If $a'_i x - b_i < 0$ for any i , then $p(x) = -\infty$. Thus the dual of (D) is

$$\begin{aligned} & \max -c'x \quad \text{or} \quad \min c'x \\ & \text{subject to } A'x \geq b, x \geq 0 \end{aligned}$$

The Lagrangian optimality condition for (P) is

$$x^* = \arg \min_{x \geq 0} \left\{ \left(c - \sum_{i=1}^m \mu_i^* a_i \right)' x + \sum_{i=1}^m \mu_i^* b_i \right\},$$

from which we determine the complementary slackness conditions for (P):

$$\begin{aligned} & \left(c_j - \sum_{i=1}^m \mu_i^* a_{ij} \right) x_j^* = 0, \quad x_j^* \geq 0, \quad \forall j = 1, \dots, n, \\ & c - \sum_{i=1}^m \mu_i^* a_i \geq 0, \quad \forall i. \end{aligned}$$

The Lagrangian optimality condition for (D) is

$$\mu^* = \arg \min_{\mu \geq 0} \{(Ax^* - b)'\mu - c'x^*\},$$

from which we determine the complementary slackness conditions for (D):

$$\begin{aligned} & Ax^* - b \geq 0, \\ & (Ax^* - b)_i \mu_i^* = 0, \quad \forall i. \end{aligned}$$

4.4.4 www

(a) Let λ_j be a Lagrange multiplier associated with the constraint $\sum_{i=1}^m x_{ij} = \beta_j$, and let ν_i be a Lagrange multiplier associated with the constraint $\sum_{j=1}^n x_{ij} = \alpha_i$. Define

$$X = \{x \mid x_{ij} \geq 0, \quad \forall i, j\}.$$

The Lagrangian function is

$$\begin{aligned} L(x, \nu, \lambda) &= \sum_{i,j} a_{ij} x_{ij} + \sum_{i=1}^m \nu_i \left(\alpha_i - \sum_{j=1}^n x_{ij} \right) + \sum_{j=1}^n \lambda_j \left(\beta_j - \sum_{i=1}^m x_{ij} \right) \\ &= \sum_{i,j} (a_{ij} - \nu_i - \lambda_j) x_{ij} + \sum_{i=1}^m \nu_i \alpha_i + \sum_{j=1}^n \lambda_j \beta_j. \end{aligned}$$

The dual function is

$$q(\nu, \lambda) = \inf_{x \in X} L(x, \nu, \lambda) = \begin{cases} \sum_{i=1}^m \nu_i \alpha_i + \sum_{j=1}^n \lambda_j \beta_j & \text{if } a_{ij} - \nu_i - \lambda_j \geq 0 \text{ for all } i, j, \\ -\infty & \text{otherwise.} \end{cases}$$

An alternative dual function is obtained by assigning a Lagrange multiplier λ_j to each constraint $\sum_{i=1}^m x_{ij} = \beta_j$, and lumping the remaining inequality constraints within the abstract set constraint. Thus,

$$X = \{x \mid \sum_{j=1}^n x_{ij} = \alpha_i, \quad x_{ij} \geq 0, \quad \forall i, j\}.$$

The Lagrangian function is

$$\begin{aligned} L(x, \lambda) &= \sum_{i,j} a_{ij} x_{ij} + \sum_{j=1}^n \lambda_j \left(\beta_j - \sum_{i=1}^m x_{ij} \right) \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n (a_{ij} - \lambda_j) x_{ij} \right) + \sum_{j=1}^n \lambda_j \beta_j. \end{aligned}$$

Then the dual function is

$$\begin{aligned} q(\lambda) &= \inf_{x \in X} L(x, \lambda) \\ &= \sum_{j=1}^n \lambda_j \beta_j + \inf_{x \in X} \sum_{i=1}^m \left(\sum_{j=1}^n (a_{ij} - \lambda_j) x_{ij} \right) \\ &= \sum_{j=1}^n \lambda_j \beta_j + \sum_{i=1}^m \inf_{1 \leq j \leq n} (a_{ij} - \lambda_j) \alpha_i, \end{aligned}$$

and the dual problem is

$$\begin{aligned} &\text{maximize } q(\lambda) \\ &\text{subject to } \lambda \in \Re^n. \end{aligned}$$

(b) & (c) The Lagrange multiplier λ_j can be interpreted as the price p_j . So if the transportation problem has an optimal solution x^* , then its dual also has an optimal solution, say p^* , and

$$q(p^*) = \sum_{i,j} a_{ij} x_{ij}^*,$$

i.e.,

$$\sum_{j=1}^n p_j^* \beta_j + \sum_{i=1}^m \min_{1 \leq j \leq n} (a_{ij} - p_j^*) \alpha_i = \sum_{i,j} a_{ij} x_{ij}^*. \quad (1)$$

Since x^* is primal feasible, we have

$$\sum_{j=1}^n p_j^* \beta_j = \sum_{j=1}^n p_j^* \sum_{i=1}^m x_{ij}^*,$$

and by combining this with Eq. (1), we obtain

$$\sum_{i=1}^m \min_{1 \leq j \leq n} \{a_{ij} - p_j^*\} \alpha_i = \sum_{i,j} (a_{ij} - p_j^*) x_{ij}^*. \quad (2)$$

By the feasibility of x^* , we have $\sum_{j=1}^n x_{ij}^* = \alpha_i$ for all i , and from Eq. (2) it follows that

$$\sum_{i,j} (a_{ij} - p_j^* - \min_{1 \leq j \leq n} \{a_{ij} - p_j^*\}) x_{ij}^* = 0.$$

Since all the terms in the summation above are nonnegative, we must have

$$\left(a_{ij} - p_j^* - \min_{1 \leq j \leq n} \{a_{ij} - p_j^*\} \right) x_{ij}^* = 0, \quad \forall i, j.$$

Therefore if $x_{ij}^* > 0$, then

$$a_{ij} - p_j^* = \min_{1 \leq k \leq n} \{a_{ik} - p_k^*\},$$

which can be equivalently expressed as

$$p_j^* - a_{ij} = \max_{1 \leq k \leq n} \{p_k^* - a_{ik}\}.$$

Since p^* is arbitrary, this property holds for every dual optimal solution p^* .

4.4.5 (Duality and Zero Sum Games) www

Consider the linear program

$$\min_{\substack{\zeta \geq A'x \\ \sum_{i=1}^n x_i = 1, x_i \geq 0}} \zeta,$$